

mutation selection

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```
library(readxl)
library(dplyr)
```

```
##
## Attaching package: 'dplyr'

## The following objects are masked from 'package:stats':
##
##   filter, lag

## The following objects are masked from 'package:base':
##
##   intersect, setdiff, setequal, union
```

```
library(tidyr)
library(ggplot2)
library(broom)
library(ggforce)
```

```
# Read mutation and clinical data
mut_df <- read_excel("41586_2021_4278_MOESM4_ESM.xlsx", sheet = 2)
clin_df <- read_excel("41586_2021_4278_MOESM4_ESM.xlsx", sheet = 1)

# Calculate mutation frequency of each gene across patients
mut_counts <- mut_df %>%
  select(Donor.ID, Hugo_Symbol) %>%
  distinct() %>%
  group_by(Hugo_Symbol) %>%
  summarise(mutated_samples = n()) %>%
  mutate(freq = mutated_samples / n_distinct(mut_df$Donor.ID)) %>%
  filter(freq > 0.05) # Keep genes mutated in >5% of samples

selected_genes <- mut_counts$Hugo_Symbol

# Create a wide-format matrix indicating whether each gene is mutated in each patient
mut_matrix <- mut_df %>%
  filter(Hugo_Symbol %in% selected_genes) %>%
  select(Donor.ID, Hugo_Symbol) %>%
  distinct() %>%
  mutate(mutated = 1) %>%
```

```

pivot_wider(names_from = Hugo_Symbol, values_from = mutated, values_fill = 0)

# Merge with clinical data
merged_df <- clin_df %>%
  select(Donor.ID, pCR.RD) %>%
  mutate(pCR = ifelse(pCR.RD == "pCR", 1, 0)) %>%
  left_join(mut_matrix, by = "Donor.ID") %>%
  replace(is.na(.), 0)

# Identify gene columns
gene_cols <- setdiff(colnames(merged_df), c("Donor.ID", "pCR.RD", "pCR"))

# Run univariable logistic regression: pCR ~ each gene
results_list <- lapply(gene_cols, function(gene) {
  model <- glm(pCR ~ merged_df[[gene]], data = merged_df, family = binomial)
  est <- summary(model)$coefficients[2, ]
  ci <- suppressMessages(confint(model))[2, ]
  data.frame(
    Gene = gene,
    OR = exp(est["Estimate"]),
    CI_lower = exp(ci[1]),
    CI_upper = exp(ci[2]),
    p_value = est["Pr(>|z|)"]
  )
})

```

```

## Warning: glm.fit: fitted probabilities numerically 0 or 1 occurred
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```

[illegible]

```

results_df <- bind_rows(results_list)

# Add labels and color categories
results_df <- results_df %>%
  mutate(
    label = paste0("OR = ", round(OR, 2),
                  "\nCI: [", round(CI_lower, 2), ", ", round(CI_upper, 2), "]",
                  "\nP = ", signif(p_value, 2)),
    color = case_when(
      p_value > 0.05 ~ "nonsignificant",
      OR >= 1 ~ "risk",
      OR < 1 ~ "protective"
    )
  )

# Paginate (10 genes per page)
genes_per_page <- 10
results_df <- results_df %>%
  arrange(desc(OR)) %>%
  mutate(page = ceiling(row_number() / genes_per_page))

n_pages <- max(results_df$page)

# Generate paginated forest plots
plots <- lapply(1:n_pages, function(p) {
  ggplot(results_df %>% filter(page == p), aes(x = reorder(Gene, OR), y = OR)) +
    geom_point(aes(color = color), size = 4) +
    geom_errorbar(aes(ymin = CI_lower, ymax = CI_upper, color = color), width = 0.2, size = 1) +
    geom_hline(yintercept = 1, linetype = "dashed", color = "gray40") +
    coord_flip() +
    scale_color_manual(values = c("risk" = "red", "protective" = "blue", "nonsignificant" = "gray60")) +
    scale_y_log10() +
    geom_text(aes(label = label), hjust = -0.1, size = 4) +
    theme_minimal(base_size = 14) +
    labs(
      y = "Odds Ratio (log scale)",
      x = NULL,
      title = paste("Forest Plot - Page", p)
    )
})

```

```

## Warning: Using 'size' aesthetic for lines was deprecated in ggplot2 3.4.0.
## i Please use 'linewidth' instead.
## This warning is displayed once every 8 hours.
## Call 'lifecycle::last_lifecycle_warnings()' to see where this warning was
## generated.

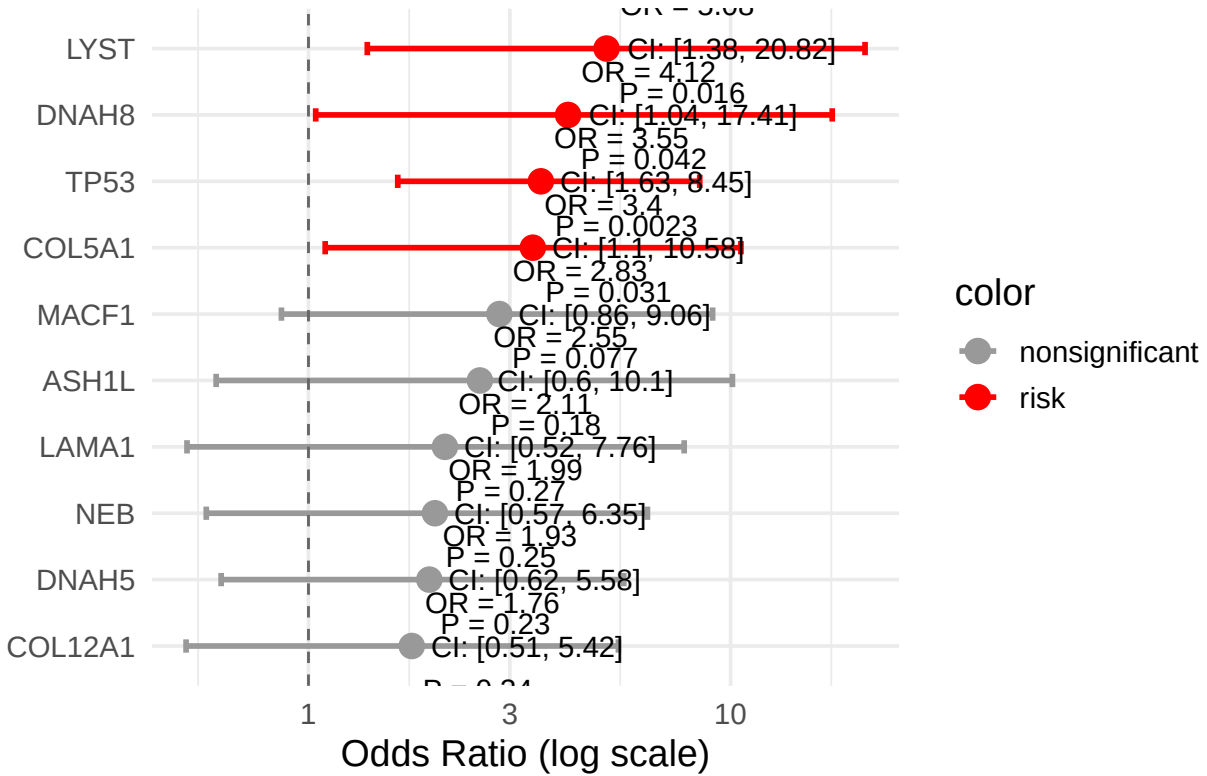
```

```

# Print each page and insert page break
for (p in 1:n_pages) {
  print(plots[[p]])
  cat("\n\nnewpage\n") # For PDF output, forces a new page
}

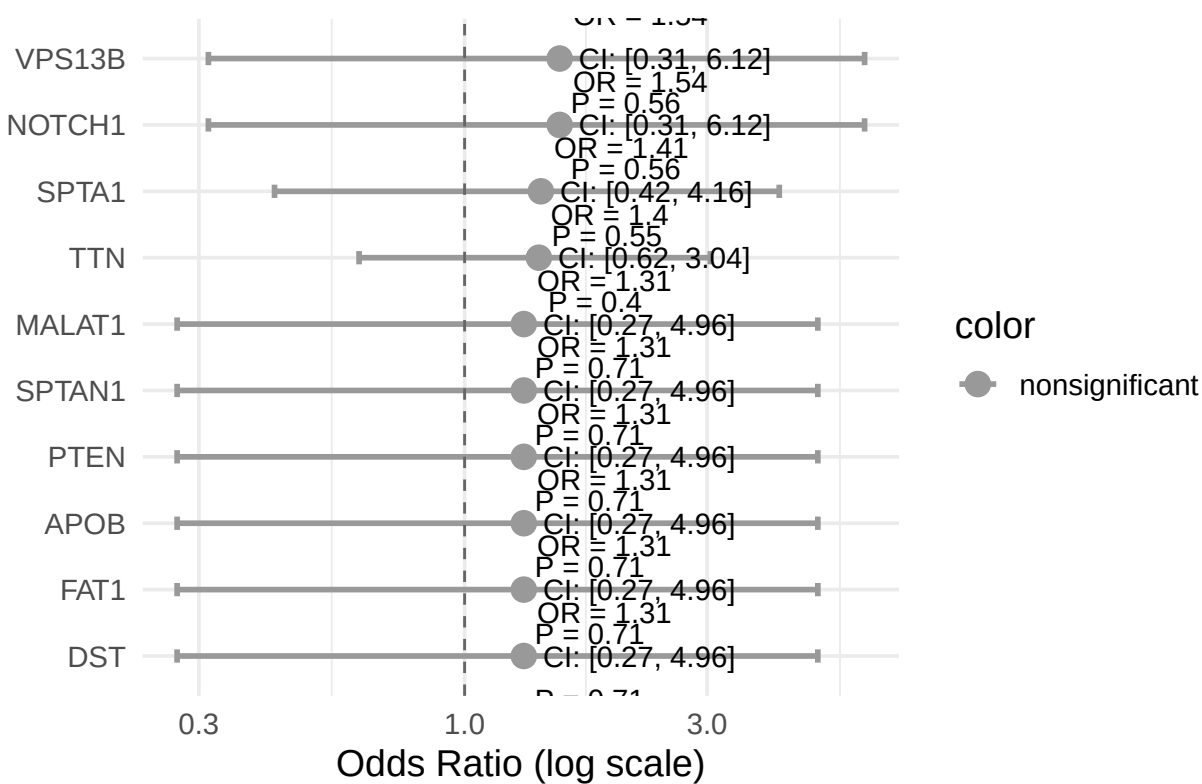
```

Forest Plot – Page 1



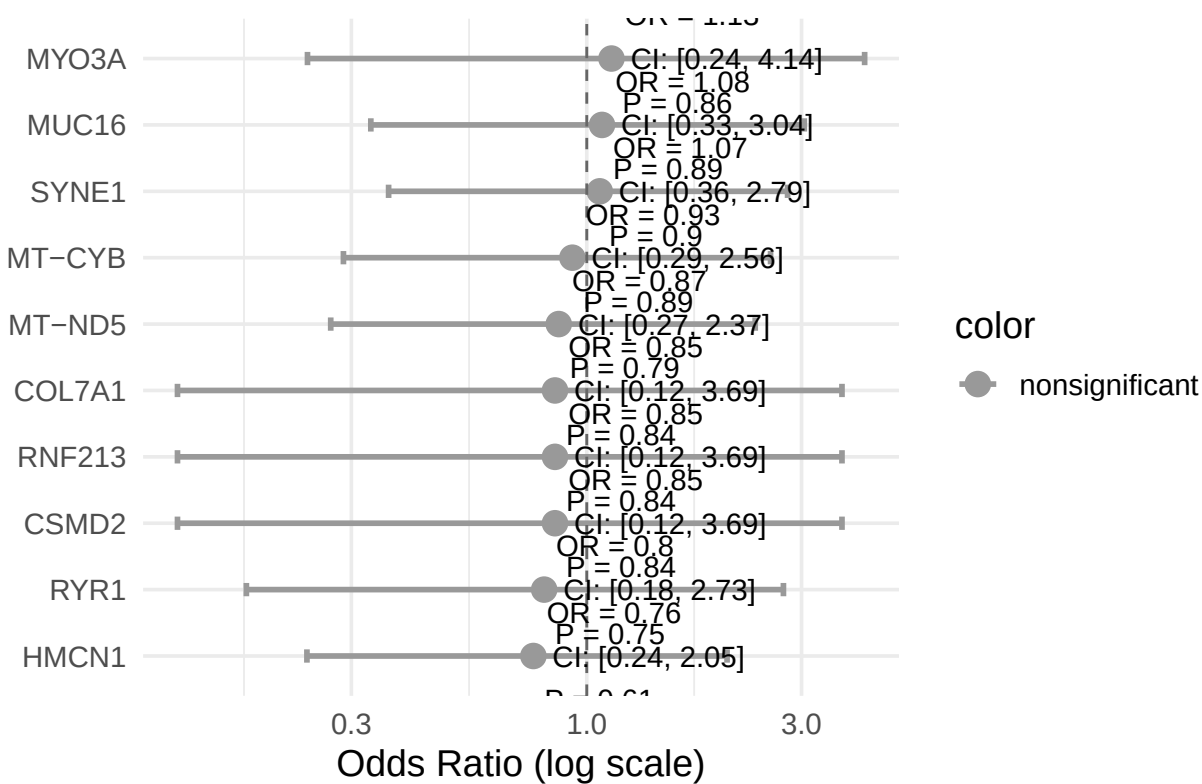
\newpage

Forest Plot – Page 2



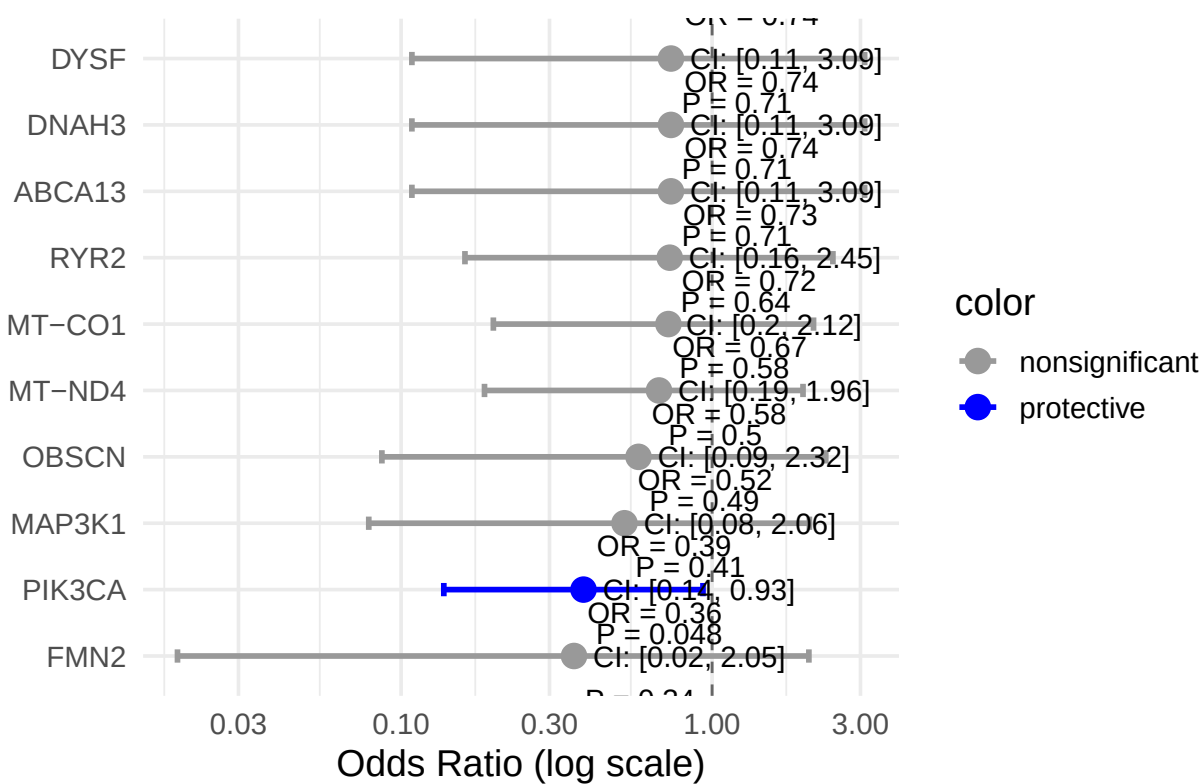
\newpage

Forest Plot – Page 3



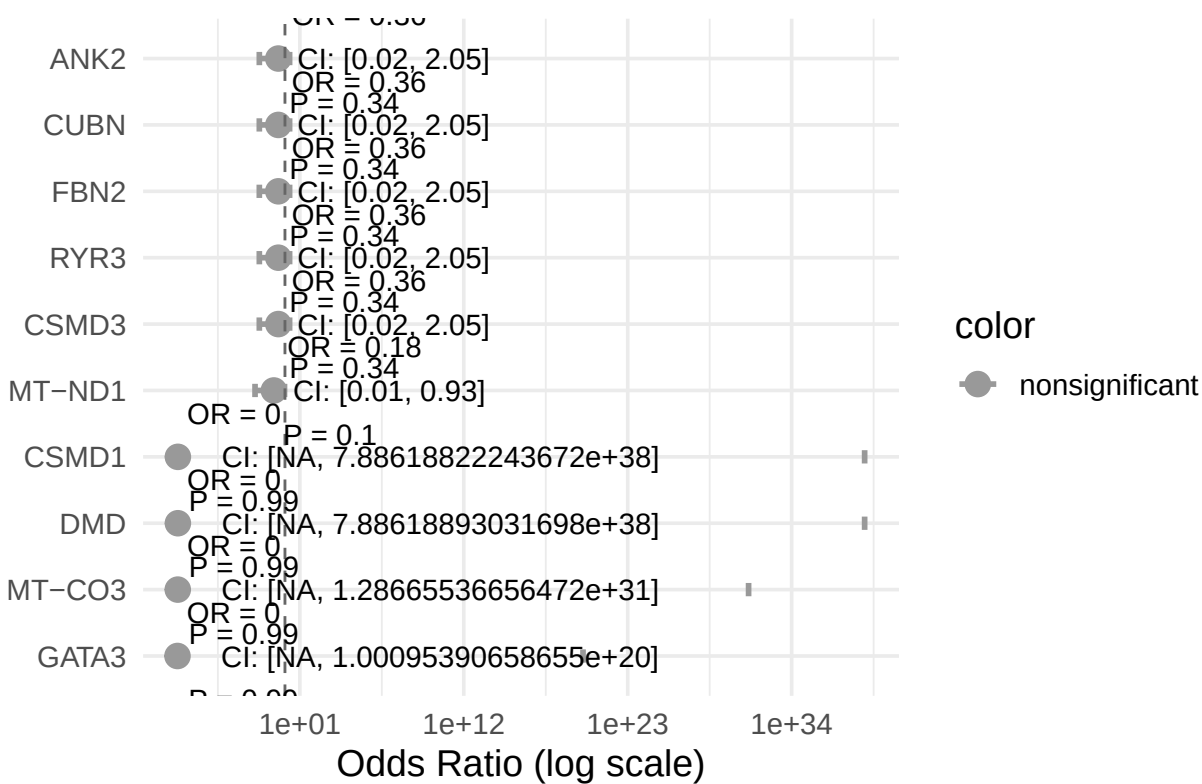
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Forest Plot – Page 4



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Forest Plot – Page 5



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